Track, AutoDesign scores 78.32; under the same Claude Code and Claude 4.8 configuration, it outperforms the closed-source commercial system Claude Design by 7.45 points. AutoDesign also makes strong paper-to-poster generation accessible at low cost: with DesignHarness, LongCat-2.0 reaches 55.13 at approximately $0.27 per poster[1]. Finally, across 933 valid system-blind pairwise judgments, AutoDesign receives the highest Bradley–Terry preference estimate, 64.0% (95% interval: 55.2–77.8%); for pairs separated by at least 20 PosterBench points, participants prefer the PosterBench-preferred poster in 74.4% of cases.

Our contributions are:

1. **AutoDesign**, a meta-harness optimization framework that turns a static design harness into a recursively improving system for human-aligned multimodal design. Through 7 days of evolving traces, it invokes 224 subagents, records at least 123 recursive iterations, accumulating 54 harness updates that recursively convert human-designed reference artifacts, rollout traces, rendering diagnostics, and evaluator feedback into persistent design priors for a design harness.
2. **DesignHarness**, the executable academic paper-to-poster system evolved by AutoDesign. It grounds a tool-using Designer in paper context and combines editable generation with rendering, rule-based validation, and visual critic feedback for localized revision, producing source-grounded posters that remain directly usable and editable.
3. **PosterBench**, a comprehensive evaluation protocol for paper-to-poster evaluation. It evaluates scientific communication quality and executable-artifact reliability through a seven-dimensional rubric spanning faithfulness, coverage, density, visual evidence, layout, readability, and aesthetics. On PosterBench, the state-of-the-art coding agent Claude Code (Claude 4.8) achieves 70.01; attaching DesignHarness can improve it by +8.31, also surpassing the best commercial design agent Claude Design by 7.45 points.
4. Under a fully autonomous long-horizon agentic loop, DesignHarness produces human-level academic posters. In a poster generation run, it executes 253 tool calls and 11 editing turns within 40 minutes for less than $3, with negligible human intervention. Demo is it available as a research-preview platform for interactive use and localized revision of editable posters.

2 Meta-Harness Formulation

2.1 Design Harness Definition

Following recent work that treats harnesses as an optimization target distinct from model weights (Lee et al., 2026b; Ren et al., 2026), we define a *design harness* H as the system surrounding a fixed model that turns a multimodal source (*e.g.*, an academic paper or report) into a human-facing artifact (*e.g.*, a presentation slide, poster, video, or web page):

$$y \sim H(\pi_\theta, x, c), \tag{1}$$

where π_θ is an LLM or MLLM, x is the multimodal input, and c specifies the context including the target medium and user constraints. The harness produces an artifact y through an execution trajectory τ, which records the sequence of intermediate actions, states, and revisions leading to the final output.

To enable systematic meta-harness optimization and facilitate credit assignment, we decompose the design harness H into five functional components:

- **Context and Memory**: source management, prompts, skills, reusable assets and persistent state.
- **Tools and Specifications**: tools and editable artifact specifications for layout, typography, and provenance.
- **Execution Runtime**: the workspace and runtime for authoring, rendering, validating, and exporting artifacts.
- **Orchestration**: task routing, attempt budgets, loop control, candidate selection, fallback, and finalization.
- **Evaluation and Feedback**: rule-based validation, model-based critique, and localized feedback for revision.

This decomposition specifies only the high-level abstraction of the design harness. The concrete implementation of each component is instantiated and iteratively improved by the meta-harness.

[1]The reported cost reflects LongCat-2.0's pricing policy at evaluation time: cached context incurs no charge on a cache hit.